

PEDALSHIELD

Ride private. Earn shielded.

A Manifesto

Why Pedalshield Will Scale the Entire Zcash System

The thesis in one line: Zcash does not have a technology problem. It has a demand problem. Pedalshield is a demand engine — a consumer app that manufactures recurring, real-world shielded transactions and turns the people who secure the chain into the people who fund it.

Prepared for prospective investors and ecosystem sponsors
June 2026 · ZecHub Hackathon 2026, Games track · Mainnet-proven

1. The bet we are making

Every serious privacy chain eventually confronts the same uncomfortable fact: cryptography is necessary but not sufficient. Zcash already has the strongest shielded technology in the industry — Orchard, Halo 2, and the current NU6.2 network rules. What it has lacked is a **steady stream of ordinary people making private payments for ordinary reasons**. Shielded adoption is the metric that matters, and it is fundamentally a **demand** problem, not a protocol problem.

Pedalshield is built to attack exactly that. It is a privacy-first, mobile bike-to-earn app: cyclists track real rides, the phone verifies them entirely on-device, and an autonomous backend pays out real shielded ZEC on mainnet. Every paid ride is a shielded Orchard transaction that would not otherwise exist. Scale the riders and you scale shielded throughput, treasury velocity, and the lived, demonstrated utility of the asset — the three things Zcash most needs and can least manufacture on its own.

This document makes the case to investors and sponsors that backing Pedalshield is not a bet on one fitness app. It is a bet on the **demand side of the entire Zcash economy**.

2. Proof, not promises

Manifestos are cheap. This one opens with receipts. The full ride loop — real outdoor ride → on-device verification → autonomous shielded payout — already runs on Zcash mainnet under current consensus, with **no human in the loop**.

What we claim	On-chain receipt
Real ride → on-device verification → autonomous payout to a rider's own wallet (Jun 11, 2026)	txid 2a849aca...b264ab — a 492 m ride, verified on the phone, paid with no operator.
Autonomous shielded Orchard payout, no operator	txid f1a3bacc...c10ba6 — fired automatically by POST /claim.
Repeatable, not a one-off	txid ef0e2a57...060587.
First autonomous spend, anchor byte-identical to GetTreeState	txid 6da9298a...272f5b74, mined in block 3,368,791.
Current with the NU6.2 emergency upgrade	Entire Zcash stack re-pinned within days; the live consensus branch id (0x5437f330) is printed by our own tooling.

Why this matters to a funder: the hardest, riskiest part of the system — a hand-rolled, fully autonomous Orchard spend pipeline built directly against the librustzcash crates with no wallet SDK — is already done and proven. We are not raising to find out whether the core works. We are raising to scale a machine that already works.

3. Why a private ride app is the right wedge

Consumer crypto has spent years searching for an app ordinary people use for non-speculative reasons. Move-to-earn got closest to mainstream — and then taught two expensive lessons. Strava proved there is a massive, devoted audience for ride tracking, and then proved, repeatedly, that centralized route data is a liability: heatmaps that de-anonymized military bases and exposed private homes. STEPN proved that rewards drive explosive growth, and then proved that rewards built on tokenomics collapse the moment yield falters.

Pedalshield is the counterexample to both, and the counterexample **is** the product. Routes are never uploaded — not encrypted-and-uploaded, never uploaded — and that non-collection is **architecturally enforced and unit-tested**, not promised in a policy. Rewards are real ZEC from a finite treasury; we never mint anything, and payouts are small and capped by design. Privacy is the product; yield is not.

The strategic point for Zcash: this is a wedge no privacy-indifferent competitor can copy without becoming us. Every Strava privacy scandal is free marketing for the only ride tracker where surveillance is impossible by construction. The privacy thesis is simultaneously the growth engine and the moat — and it runs natively on Zcash because Zcash is the only chain whose shielded payments make the story true end to end.

4. How the architecture scales the chain

A skeptic's first objection is throughput: one on-chain transaction per ride does not scale. Correct — and it does not need to. The path to millions of rides is **accrual plus batched settlement**, and it is already designed:

- **Accrue, don't pay per ride.** Verified kilometers credit a rider balance; payout settles monthly or at a rider-set threshold.
- **Batch into multi-output Orchard transactions.** One shielded tx carries ~50 recipients. ZIP-317 fees stay trivial — about 0.0025 ZEC per 50-payout tx, roughly \$0.02 per rider per settlement.
- **Pre-split note pool.** The treasury keeps a pool of pre-split notes so settlement transactions never contend for the same note, killing the double-select race and enabling parallel worker treasuries.

The capacity math is comfortable for the chain, by design:

Volume	Settlement txs	Daily load on Zcash
1M rides / month (~100K riders)	~2,000 txs / month	~65 txs / day
10M rides / month (~1M riders)	~20,000 txs / month	~650 txs / day

The honest read: chain throughput is never the bottleneck — treasury funding is. That is exactly the right problem to have, because it is one investors and sponsors can solve, and solving it routes capital and mining yield into shielded usage. Verification stays fully on-device at any scale (zero marginal server cost), and the spend path is hand-rolled with no SDK lock-in, so the inevitable Orchard→Ironwood migration is a budgeted annual cost, not an existential risk.

5. The flywheel: turning miners into the supply side

Here is where Pedalshield stops being an app and starts being Zcash infrastructure. Scaling to millions of private rides needs three scarce things: **always-on infrastructure, distributed custody of a real treasury, and ZEC to pay riders**. Zcash miners are uniquely matched to supply all three at once. They already hold ZEC and want utility for it beyond dumping block rewards into the market. They already run 24/7 infrastructure with real key hygiene. And as block rewards tighten across halvings, they are actively hunting for protocol-aligned yield — precisely what Pedalshield offers.

So miners are not one audience but a pipeline through four roles, from lowest commitment to highest:

Role	What they supply	What they get
Node / settlement operator	Compute + uptime; run lightwalletd, settlement workers, ledger replicas (no spend authority)	Per-settlement infra fee; public leaderboard; first refusal on signer seats
Treasury steward / FROST signer	Custody + availability; hold one share of the threshold key and co-sign batches	Signer stipend; governance voice
Capital / liquidity provider	ZEC into the reward pool	Defined, non-yield stake (needs counsel before it ships)
Rider-growth partner	Community + onboarding	Referral rewards; cosmetic perks

Stand back and the flywheel is visible. **Miners fund and operate the treasury → the treasury pays riders in shielded ZEC → riders generate recurring shielded demand and visible utility → that utility supports the value of the very ZEC miners produce → stronger miner economics fund more rides.** “Mining funds riding” is a uniquely Zcash-native narrative, and it converts the security budget of the chain into a growth budget for shielded adoption. No other privacy chain has both the shielded primitive and the aligned, infrastructure-rich miner base to close this loop.

6. The economics hold up

A demand engine that bleeds the treasury dry is a fad. Pedalshield is engineered so reward emission tapers roughly 20× between launch and the 1M-rides tier, on a schedule published up front — Sweatcoin-style credibility, no rug. Reward rates are environment knobs, repriced without redeloys, so a volatile ZEC price changes reward attractiveness, never solvency (caps are tight and liabilities are small). All figures at ~\$426/ZEC; every number is parameterized.

Tier	Riders	ZEC / km	Treasury burn/mo	Sub revenue/mo
10K rides/mo	1K	0.0002	18 ZEC (\$7.7K)	\$0.3K
100K rides/mo	10K	0.00005	45 ZEC (\$19K)	\$3K
1M rides/mo	100K	0.00002	180 ZEC (\$77K)	\$30K
10M rides/mo	1M	0.00001	900 ZEC (\$383K)	\$300K

Settlement fees are noise (≤ 50 ZEC/mo even at 10M rides). Premium subscriptions at 5% conversion (\$5.99/mo) are one revenue line; the design target is for reward burn to trend toward $\leq 100\%$ of recurring revenue by the 1M tier, with the gap closed by sponsor-funded reward pools, corporate-wellness seats, and the miner-directed treasury experiment.

7. Why this scales the entire system, not just an app

Pull the threads together and Pedalshield touches every layer of the Zcash economy at once:

- **Usage layer** — recurring, real-world shielded transactions from non-crypto-native users — the exact adoption metric Zcash is judged on.
- **Throughput layer** — batched Orchard settlement is a neighborly, sustainable, high-volume on-chain workload that exercises and validates the shielded pool at scale.
- **Security / mining layer** — miners gain protocol-aligned utility and yield for their ZEC and infrastructure as block rewards taper, strengthening the security budget.
- **Tooling layer** — a hand-rolled, SDK-free, NU6.2-current spend pipeline and a public FROST treasury ceremony are reusable open-source infrastructure the whole ecosystem can learn from.
- **Narrative layer** — a consumer app the community can be proud of — privacy that protects real people on real streets — answering “what is shielded money actually for?” with a daily, tangible answer.

That is the difference between a project that **uses** Zcash and one that **scales** Zcash. Pedalshield manufactures demand, routes capital and mining power toward shielded usage, stress-tests the pool with sustainable volume, and gives the ecosystem a story that recruits the next builder. The flywheel compounds: more riders → more shielded demand → stronger ZEC utility → more aligned miners → more treasury capacity → more riders.

8. The constraints are the moat

Credibility, especially in privacy, is built on what you **refuse** to do. At any scale, Pedalshield will never run a route database, never sell data to cities or insurers or advertisers, never launch a token, and never make a yield promise. Growth that requires breaking these is growth we decline; the constraint is the brand, and it compounds with every competitor data scandal. We also state our limits out loud: the anti-cheat is layered, not perfect; FROST does not yet authorize mainnet Orchard spends (the RedPallas/ZIP-312 swap is on the roadmap); and ZK proof-of-distance is the trustless v2, not the present. Funders deserve a team that names its uncertainties before being asked.

9. The ask

We are raising a seed round (target ~\$1.5–2.5M) for roughly 18 months of runway: a small team (two mobile, one protocol/Rust, one growth, founder), modest treasury seeding (phase 1–2 burn is small), and a city-by-city go-to-market down the privacy×cycling wedge — Germany and the Netherlands first, then privacy-conscious US metros. Milestones investors can hold us to: the Ironwood pool migration, ≥20% D30 retention in closed beta, and 10K active riders. We raise against retention and revenue per rider — never against token mechanics.

The invitation. Back the demand side of Zcash. Fund the engine that turns ordinary bike rides into shielded transactions, turns the miners who secure the chain into the network that scales it, and turns privacy from a feature into a habit. Pedalshield is already on mainnet. The only question left is how fast it grows.

Get Pedalshield

Join the beta waitlist: pedalshield.app

For developers — try it today: github.com/intelligrip/Pedalshield

Pedalshield runs today as a developer build on the live Zcash network; the public beta opens in 2026. Join the waitlist to be notified at launch.

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